

NetSecure – Network Security Assessment Tool

Introduction

NetSecure is a beginner-friendly cybersecurity project that helps users perform a basic security assessment of authorized systems or local networks. The application identifies active devices, scans common open ports using Nmap, displays basic service information, and generates a simple security report with recommendations. It is designed to help users understand the basics of network security in a simple and practical way. By presenting scan results in an easy-to-read format, the project makes it easier to identify common security risks and encourages safe network practices. The project is intended only for educational purposes and will be used only on systems where permission has been granted.

Problem Statement

Many users are unaware of open ports and services running on their systems, which can become security risks. Existing security tools are often difficult for beginners to understand. NetSecure aims to provide a simple interface that makes basic network assessment easier.

Objectives

- Discover active devices on an authorized network.
- Scan common open ports using Nmap.
- Display basic service information.
- Classify the risk level as Low, Medium, or High.
- Generate a simple security report with recommendations.

Proposed Solution

The user enters an IP address or domain name. The application performs an authorized scan, collects information about open ports and services, analyzes the results using predefined rules, and displays them in a simple dashboard. A report is then generated for future reference.

Modules

- **Scanning Module** – takes the IP address or domain, resolves it if needed, and runs the Nmap scan to find open ports and services.
- **Risk Analysis Module** – takes the scan output and classifies the risk as Low, Medium, or High based on predefined rules.
- **Logging Module** – saves every scan into the SQLite database and lets the user view or search past logs.
- **Report and Dashboard Module** – the Flask web pages that show the scan results on a dashboard and generate the final report.

Methodology / Working

- 1 Enter an IP address or domain.

- 2 Resolve the domain to an IP address if required.
- 3 Perform an authorized Nmap scan.
- 4 Display open ports and detected services.
- 5 Assign a basic risk level.
- 6 Store the scan result in the database using SQL.
- 7 Generate a security report with recommendations.

Tools, Languages and Frameworks Used

Language: Python

Framework: Flask

Frontend: HTML, CSS, Bootstrap

Tools: Nmap, Wireshark (for packet observation)

Database: SQLite, used to store scan logs

Expected Outcome

The application will help users perform a basic authorized network assessment and understand common network services, open ports, and simple cybersecurity practices through an easy-to-use interface. It will also keep a simple log of past scans so users can look back at earlier results.

Future Scope

Future versions can include improved reporting, graphical dashboards, user authentication, and additional security checks.